REMEMBERING WHERE TO LOOK: CONTINUAL HIERARCHICAL EVIDENCE SELECTION FOR WHOLE-SLIDE IMAGES

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ABSTRACT

Whole-slide image (WSI) diagnosis selects a few evidence patches among thousands. Continual-learning methods preserve classifiers or representations, but a budgeted system can fail differently: its diagnostic pathway may remain intact while its selector forgets where to look. **PathSelect** makes the evidence-selection policy itself the continual-learning object. The CONCH image and text encoders and the cosine diagnostic head are frozen, isolating forgetting to a hierarchical group-to-patch selector. Diagnostic, semantic, and counterfactual-utility supervision teach the selector where to look within a task; across tasks, replay and dual-level distillation preserve prior behavior, a utility-preservation hinge lets evidence change as long as its diagnostic usefulness does not regress, and task-local LoRA updates fold into one shared task-free selector. On the four-cohort ConSlide benchmark it reaches 0.854 class-incremental accuracy with 0.060 forgetting, competitive with the strongest published result (0.859), versus 0.452 and 0.601 without preservation; selection-level measurements show that evidence identity and utility drift differently.

Index Terms— whole-slide images, continual learning, evidence selection, computational pathology, LoRA

1. INTRODUCTION

A gigapixel whole-slide image (WSI) may contain tens of thousands of patches, yet diagnosis often depends on only a small subset. Practical systems therefore operate under an *evidence budget*: select a few patches, then decide. This makes evidence acquisition itself a potential locus of catastrophic forgetting. A model may retain its visual representation and diagnostic rule after learning new disease cohorts, yet still fail because its selector no longer retrieves the tissue that previously supported the correct decision. Conventional continual learning, which primarily measures preservation of predictions, representations, or classifier parameters, does not directly expose this failure.

We therefore ask: when diagnostic tasks arrive sequentially, does a model forget where to look, and can that ability be measured and preserved directly? PathSelect is built around this question. To isolate selector forgetting, we freeze the CONCH image and text encoders [1] and the cosine diagnostic rule throughout training; the evidence-selection policy is the only learned continual object. This controlled setting lets us study changes in evidence acquisition directly rather than infer them from downstream accuracy. The selector follows the tissue-aware group-to-patch hierarchy of HistoSelect [2]; our focus is not the hierarchy itself but whether a learned selection policy remains competent as new diagnostic tasks arrive.

This leads to a deliberate separation between learning and preservation. Within a task, PathSelect must *learn where to look now*, using diagnostic, semantic, and counterfactual-utility supervision. Across tasks, it must *keep looking well*. We distinguish *behavioral continuity*—whether the selector retains its old scoring preferences—from *functional continuity*—whether the evidence it now selects remains diagnostically useful. The distinction matters because changing the selected patches is not necessarily forgetting: alternative evidence should be allowed if it is at least as useful as what was selected before. PathSelect therefore combines dual-level distillation, utility preservation, and replay, while task-local LoRA updates are folded into a single shared selector for task-free inference.

Contributions.

- We formulate continual WSI diagnosis as continual evidence selection: with the image encoder, text encoder, and diagnostic head frozen, the hard budgeted selector is the only learned, preserved, and directly evaluated continual object. This isolates a failure mode largely hidden in prior WSI continual learning—forgetting where to look even when the diagnostic pathway itself is unchanged.
- We introduce PathSelect, a fixed-cost continual selector that preserves both selection behavior and selection usefulness. Dual-level distillation stabilizes old group- and patch-level scoring, while a utility-preservation hinge allows evidence identity to change as long as diagnostic usefulness does not regress. Per-task LoRA updates are folded into one shared task-free selector, with replay driven by a compact behavioral memory that stores selector snapshots rather than image or feature tensors.
- We directly measure two complementary forms of selection drift—evidence identity with Jaccard and diagnostic function with utility retention—and show that they are not interchangeable: preserving where a selector looks does not guarantee preserving how useful what it finds remains. On the ConSlide four-cohort benchmark, PathSelect reaches 0.854 ACC, competitive with the strongest published continual result (0.859), while substantially reducing selector forgetting relative to unprotected sequential learning.

2. RELATED WORK

Evidence selection and continual WSI learning. Classical MIL methods such as ABMIL, CLAM, and TransMIL [3, 4, 5] aggregate patch features with learned attention, whereas HistoSelect [2] makes evidence selection explicitly tissue-aware and hierarchical, conditioned on a per-question query; recent active-perception methods

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learn where to inspect in a WSI [6] but do not address continual retention. In parallel, continual WSI methods preserve diagnostic competence through rehearsal, attention or aggregation regularization, prototypes, or model adaptation: ConSlide [7] introduced rehearsal-based continual WSI analysis on a four-cohort TCGA benchmark; MICIL [8] studied class-incremental MIL; QPMIL-VL [9] used vision–language prototypes on CONCH features; Li et al. [10] distilled MIL attention with a pseudo-bag memory; and CoMEL [11] and MergeSlide [12] explored low-rank adaptation and model merging. These methods primarily adapt or preserve learned diagnostic, aggregation, attention, or prototype components rather than isolate a standalone hard evidence selector.

PathSelect bridges these two lines. It adopts the HistoSelect group-to-patch substrate with a fixed, task-agnostic tissue vocabulary that provides a stable grouping reference across tasks, freezes the image and text encoders and the diagnostic rule, and makes the budgeted selector the sole learned continual object. Unlike methods that infer forgetting only from downstream accuracy, it measures evidence-identity and diagnostic-utility drift directly and preserves both selection behavior and usefulness. LoRA [13] and its continual-learning variants [14, 15, 16, 17] are used only for fixed-cost consolidation into one shared selector; forgetting is addressed separately through replay, dual-level distillation, and utility preservation.

3. METHOD

3.1. Study object: a frozen diagnostic pathway

Tasks $\tau = 1, \ldots, T$ arrive sequentially. A slide is represented by frozen CONCH patch features $\{x_i\}_{i=1}^{N}$, $x_i \in \mathbb{R}^{512}$, from which the selector chooses an evidence set $\mathcal{P}$ of fixed size $B = 8$. Only the selected set is used for the final diagnostic prediction; during training, additional candidate evaluations serve only to construct the counterfactual supervision of Sec. 3.3. With selector scores s_i for the chosen patches, the slide representation is $z = \sum_{i\in\mathcal{P}} \mathrm{softmax}(s)_i\, x_i \,/\, \left\|\sum_{i\in\mathcal{P}} \mathrm{softmax}(s)_i\, x_i\right\|_2$, and the prediction is the softmax over cosine similarities between z and the frozen class-text embeddings $\{t_c\}$. The embeddings of all $C = 8$ benchmark classes are computed once and form the diagnostic logits throughout training and evaluation (Sec. 4); training uses the labels of the current task. No task identity, adapter bank, memory lookup, or trained diagnostic head is used at test time.

3.2. Hierarchical budgeted selection

Following the tissue-aware coarse-to-fine principle of HistoSelect [2], patches are assigned to groups semantically rather than clustered. Eight fixed tissue prompts are encoded once by the frozen text encoder into $\{p_k\}$; each patch joins the group G_j with $j = \arg\max_k \cos(x_i, p_k)$, and each group is summarized by its prototype $g_j = \mathrm{mean}(x_i \in G_j)$. Two independent two-layer MLPs, the group selector F_g and the patch selector F_p, score groups and patches, respectively. F_g scores the prototypes, $r_j = F_g(g_j)$, and a largest-remainder allocation converts the softmax over the scores of the non-empty groups with remaining capacity into integer quotas $b_j \in \mathbb{N}$ with $\sum_j b_j = B$; other groups receive no quota. F_p scores the members of each group, $s_i = F_p(x_i)$, and the top b_j patches of each group form $\mathcal{P}$. The hierarchy thus answers two questions in turn: *which tissue groups deserve the budget*, and *which patches within those groups deserve the slots*.

3.3. Learning where to look: three within-task signals

The within-task objective $\mathcal{L}_{\mathrm{ev}} = \mathcal{L}_{\mathrm{diag}} + \beta_s \mathcal{L}_{\mathrm{sem}} + \beta_u \mathcal{L}_{\mathrm{util}}$ has three terms that supervise the selector at different levels.

Outcome supervision: $\mathcal{L}_{\mathrm{diag}}$. Cross-entropy on the frozen diagnostic logits asks whether the final evidence set answered correctly; it supervises the joint outcome but assigns credit weakly among the many candidate patches.

Semantic supervision: $\mathcal{L}_{\mathrm{sem}}$. A frozen class-text prior scores how diagnostically discriminative each candidate patch appears in the CONCH embedding space: with $z_{ic} = \cos(x_i, t_c)/T$, the prior is $p_i \propto 1 - H(\mathrm{softmax}_c(z_{ic}))/\log C$, and the selector's score distribution over candidates is softly aligned to it with a KL term. Class-derived information thus enters training through this prior and through the frozen-head diagnostic losses used for $\mathcal{L}_{\mathrm{diag}}$ and for the counterfactual teacher below; none of it is provided to the selector as input at inference. The prior is a weak semantic anchor, not the diagnostic target.

Counterfactual credit: $\mathcal{L}_{\mathrm{util}}$. Let E denote the currently selected set, summarized by the equal-weight mean of its patches (a pre-registered approximation of the head's pooling). For each unselected candidate i among the top-256 scored patches, the marginal gain

$$u_i = \mathcal{L}_{\mathrm{diag}}(E) - \mathcal{L}_{\mathrm{diag}}(E \cup \{x_i\}) \tag{1}$$

is positive when adding the candidate would reduce the diagnostic loss; it is computed by a rank-one update of the pooled representation, used only to build a no-gradient teacher, and $\mathcal{L}_{\mathrm{util}} = \mathrm{KL}(\mathrm{softmax}(u) \,\|\, \mathrm{softmax}(s))$ trains the patch selector to rank candidates by measured usefulness. The lowercase u_i is a *candidate-level* signal; the value of the *whole selected set*, $U(\mathcal{P}) = \log C - \mathrm{CE}(\mathcal{P}, y)$, measures how much the set improves over a uniform diagnostic state—a distinction that becomes central across tasks.

3.4. One shared selector: LoRA consolidation

For task τ, low-rank LoRA adapters are attached to the four linear layers of F_g and F_p; the base selector is frozen during the task and only the current adapters receive gradients. At the task boundary the residual of each adapted layer ℓ is folded into its base weight, $W_\tau^{(\ell)} = W_{\tau-1}^{(\ell)} + B_\tau^{(\ell)} A_\tau^{(\ell)}$, an exact rewrite of the effective weight used immediately before, and the adapters are reset for the next task. The parameter count therefore does not grow with T, and inference always uses one shared selector. LoRA consolidation provides a fixed-cost shared-selector parameterization; continual preservation is handled separately by the objectives of Sec. 3.5.

3.5. Keeping looking well: preserving behavior and function

After a task is learned, PathSelect stores a bounded Selection Memory $\mathcal{M}$ of at most 512 behavioral snapshots, filled by reservoir sampling. A snapshot stores an internal sample reference, the task tag, group scores r_{old} (8 values), candidate indices ($\leq$256), patch scores s_{old} ($\leq$256), and utilities ($\leq$256); it contains no raw image or feature tensor, and the 512-entry memory occupies 3.11 MB when serialized. During later tasks one snapshot is replayed per training step: its slide's frozen features are reloaded from the training feature store by the sample reference and pass through the same forward path, so replay assumes authorized training-time access to the feature records (19.5 GB here), which the memory figure excludes. The continual

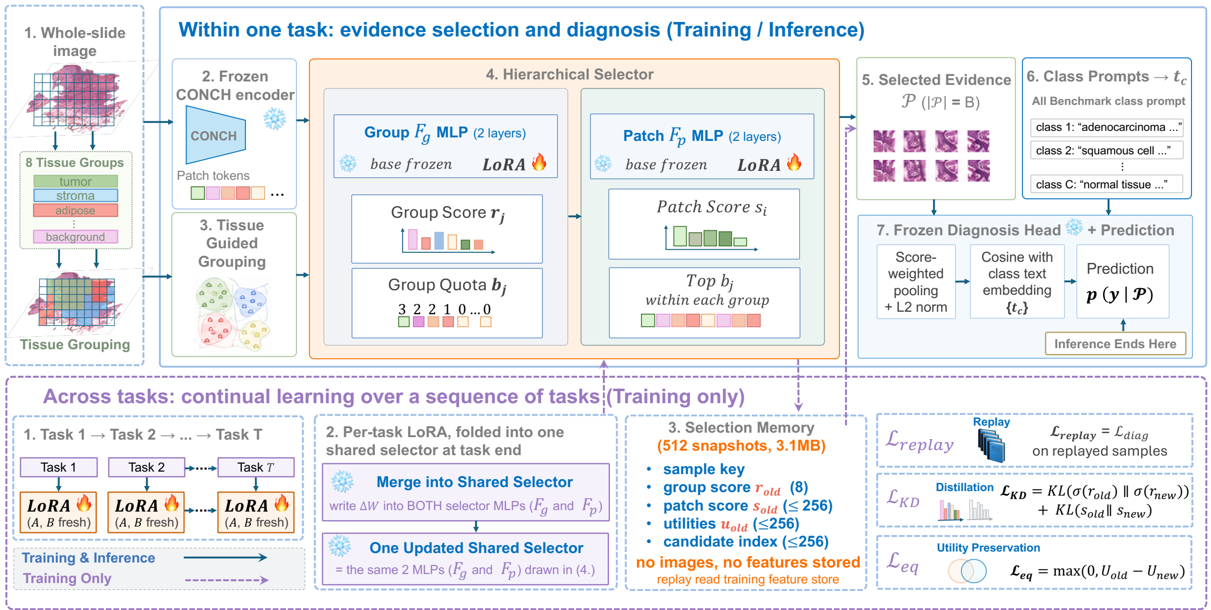

Fig. 1. **PathSelect overview.** Top (solid; training and inference): frozen CONCH features are grouped by eight fixed tissue prompts, F_g turns group scores into integer quotas ($\sum_j b_j = B = 8$), F_p keeps the top b_j patches per group, and the frozen head diagnoses from the selected evidence by cosine with class-text embeddings—inference ends there. Bottom (dashed; training only): per-task LoRA adapters are folded into one shared selector, and a Selection Memory of 512 behavioral snapshots (no images or features) drives replay, dual-level distillation, and the utility-preservation hinge.

objective is

$$\mathcal{L}_{\mathrm{CL}} = \lambda_{\mathrm{kd}} \left[\mathrm{KL}\big(\sigma(r_{\mathrm{old}}) \| \sigma(r_{\mathrm{new}})\big) + \mathrm{KL}\big(\sigma(s_{\mathrm{old}}) \| \sigma(s_{\mathrm{new}})\big)\right] + \lambda_{\mathrm{eq}} \max(0, U_{\mathrm{old}} - U_{\mathrm{new}}) + \lambda_r \mathcal{L}^{\mathrm{rep}}_{\mathrm{diag}}, \quad (2)$$

where σ denotes the softmax over groups and over candidates, respectively; its three terms play complementary roles. **Behavioral continuity** ($\mathcal{L}_{\mathrm{KD}}$): dual-level distillation preserves the old *soft* selection policy—group preferences and patch-score distributions—without requiring the hard top-B set to be identical. **Functional continuity** ($\mathcal{L}_{\mathrm{eq}}$): the hinge is a *floor, not a ceiling*—regression is penalized, improvement is free—so identity may change if the alternative evidence is at least as useful; this is the difference between remembering the same evidence and retaining the same selection competence. **Task competence**: replay applies the ordinary diagnostic loss $\mathcal{L}^{\mathrm{rep}}_{\mathrm{diag}}$ to replayed old slides, with no replay-specific loss. The task objective $\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{ev}} + \mathcal{L}_{\mathrm{CL}}$ is optimized over the current task's LoRA adapters.

4. EXPERIMENTS

4.1. Protocol and measurements

Benchmark and protocols. We use the four-cohort TCGA benchmark introduced by ConSlide [7] in the form released by QPMIL-VL [9]: CLAM-tiled 256 × 256 patches at 10×, pre-extracted CONCH features, and ten patient-level folds. Tasks arrive in the reverse order ESCA→RCC→BRCA→LUNG (LUNG denotes the NSCLC subtype task; the forward order is also reported), each a binary subtype task, for a final eight-class label space. All runs use 5 epochs per task, learning rate 10^{-3}, $B = 8$ (the operating point of the preliminary budget sweep in Sec. 4.3), LoRA rank 4, β_s=β_u=0.1, and unit λ's. The *comparison protocol* follows the benchmark's CONCH-feature release [9]: ten folds, one run per fold with the seed equal to the fold index, and the metrics average accuracy (ACC), masked ACC (task-IL), forgetting [18], and backward transfer (BWT) [19]. The *ablation protocol* fixes fold 1 (2,273 training slides; test sizes 15/76/93/95) and varies five seeds; comparisons are paired by seed with win counts, 5/5 is called systematic and 4/5 directional, and no p-values are reported at this seed count; ablation decisions followed two-sided criteria fixed before the runs, the existing design is retained on a split, and one pre-specified expectation that the data refuted is reported as measured. The controlled variants are sequential fine-tuning of the selector (SeqFT), LoRA with boundary merging only, replay, replay+distillation, and full PathSelect (replay+distillation+utility hinge).

Measurements. Across our controlled variants the eight-way class-text head is fixed at every stage, so differences arise from selector updates and incoming task data rather than from head updates. Class-IL accuracy takes the argmax of this head at every stage, so predictions may fall on classes not yet seen; restricting the argmax to the classes seen so far, as accumulating heads do, leaves final accuracy unchanged and changes forgetting by at most 0.007 in our ten-fold runs, indicating that the head convention has little effect on the reported comparison. Task-IL accuracy restricts the argmax to the task's own pair; leakage is the fraction predicted into another task's class set. Selection is measured separately: selection Jaccard, $|\mathcal{P}_{\mathrm{own}} \cap \mathcal{P}_{\mathrm{final}}|/|\mathcal{P}_{\mathrm{own}} \cup \mathcal{P}_{\mathrm{final}}|$, is the hard identity retention between the set selected immediately after learning a task and the set selected at the end (an evaluation quantity, not the distillation loss), and functional retention is $\Delta U = U_{\mathrm{final}} - U_{\mathrm{own}}$.

Table 1. **Comparison on the ConSlide four-cohort benchmark** [7] (CONCH features, reverse order ESCA→RCC→BRCA→LUNG, ten-fold CV, mean±sd). Published rows are quoted from Table 2 of [9] and were not rerun by us; PathSelect uses the identical public splits and metric definitions, while training settings follow each method. PathSelect adapts only the evidence selector over a frozen pathway (a cosine head fixed to all eight class prompts from the first task), so parameter counts characterize adaptation cost rather than capacity-matched architectures. Replay memory is in stored WSIs for published rows; PathSelect's memory holds 512 behavioral snapshots (3.1 MB of metadata, excluding the feature store required for replay). Forward-order results are in Table 2. ‡Our released-code reproduction matches the reported results within 0.01 in both orders; settings-level details (class-ensemble ordering, mini-batch, hardware) are documented in the released repository.

Type	Method	ACC↑	Forgetting↓	BWT↑	Masked ACC↑	Replay memory
Published methods—values quoted from [9]						
Reference	JointTrain (offline reference)	0.908±.022	—	—	0.937±.022	—
Reference	FineTune (lower)	0.234±.008	0.927±.023	−0.927±.023	0.803±.060	0
Regularization	EWC [20]	0.235±.011	0.928±.020	−0.928±.020	0.833±.069	0
Regularization	LwF [21]	0.236±.016	0.908±.030	−0.908±.030	0.900±.041	0
Rehearsal	A-GEM [22]	0.536±.047	0.527±.067	−0.527±.067	0.872±.026	30
Rehearsal	ER-ACE [23]	0.703±.049	0.281±.062	−0.279±.063	0.889±.041	30
Rehearsal	DER++ [24]	0.684±.055	0.310±.069	−0.307±.072	0.910±.043	30
Rehearsal	ER [25]	0.644±.028	0.387±.050	−0.386±.049	0.901±.035	30
WSI CL	ConSlide [7]	0.499±.025	0.058±.032	−0.021±.039	0.854±.039	30
VLM	AttriCLIP [26]	0.694±.058	0.207±.063	−0.207±.063	0.861±.019	0
VLM	MI-Zero [27]	0.839±.034	—	—	0.909±.019	0
VLM WSI CL	QPMIL-VL [9]	0.859±.032	0.064±.031	−0.064±.031	0.925±.018	0
Reproduction check—our run of the released code on the same folds‡						
VLM WSI CL	QPMIL-VL (reproduced)	0.868±.046	0.064±.054	−0.064±.054	0.935±.018	0
Our method—run under the same splits and metrics						
Selection CL	PathSelect (ours)	0.854±.044	0.060±.041	−0.056±.038	0.925±.019	512 snap. (3.1 MB)

4.2. Where does PathSelect stand among published methods?

Table 1 places PathSelect in the continual WSI landscape on the ConSlide four-cohort benchmark; quoting published entries follows the practice of the benchmark's own tables. PathSelect reaches $0.854{\pm}.044$ ACC, $0.925{\pm}.019$ masked ACC, and $0.060{\pm}.041$ forgetting: the second-highest ACC among the continual methods in the table, 0.005 below the strongest published result (0.859), and above every regularization, rehearsal, ConSlide, and prompt-based row; the offline joint-training reference reaches 0.908. ConSlide reports slightly lower forgetting (0.058 vs. 0.060) at substantially lower ACC (0.499), so we do not claim the lowest forgetting. Rerunning the strongest method with its released code and the paper's settings reproduces its published results in both orders to within 0.01 (reverse $0.868{\pm}.046$, forward $0.884{\pm}.033$), which cross-validates the data, folds, metric definitions, and procedure; relative to the published and reproduced reverse values, PathSelect is 0.005 and 0.014 lower in ACC, within one standard deviation of either, with comparable forgetting (0.060 vs. 0.064). We therefore describe PathSelect as competitive with the strongest published method rather than above or below it. What differs is the adaptation cost and the inference structure: PathSelect adapts 16.4k selector parameters per task, versus 0.365M learnable parameters reported for the strongest method ($22.3\times$ fewer; counts refer to different adapted components), with no prototype pool or task-specific module at inference and a 3.1 MB behavioral memory (a 30-WSI feature buffer is about 180 MB at 6.1 MB per slide).

4.3. Ablations

Which part matters, and does it hold in both task orders? (Table 2) A preliminary budget sweep with a learned flat selector (inference only) beat random selection in all 28 task×budget cells and peaked at $B = 8$ (0.8797 vs. 0.6607). Table 2 then removes one part of PathSelect at a time under the comparison protocol. On the flat substrate, removing continual preservation altogether drops ACC from 0.830 to 0.452 and raises forgetting from 0.090 to 0.601, in every fold on every metric. Selector-only fine-tuning nevertheless remains above the published full-model fine-tuning reference (0.452 vs. 0.234); because the implementations are not controlled, we treat this comparison as context rather than attribution. Replay recovers most of the accuracy (0.821); distillation and the utility hinge then add 0.8 ACC points (7/10 folds) and remove 2.4 forgetting points (8/10); the hierarchy adds 2.5 ACC points (8/10) and removes 2.9 forgetting points (7/10) under the canonical reverse order. Under the forward order the chain is $0.529 \rightarrow 0.841$ (replay) $\rightarrow 0.841$ (full, flat) $\rightarrow 0.811$ (full, hierarchical): preservation is again decisive (10/10 folds on every metric), distillation and the hinge add no accuracy (5/10) but still reduce forgetting (-0.9 points, 8/10), and the hierarchy costs 3.0 points (9/10). Across both orders the robust findings are the preservation framework itself and the forgetting reduction from distillation and utility preservation; the accuracy increments and the substrate's benefit are order-dependent, and both

Table 2. **Leave-out chain in both task orders** (ConSlide benchmark, ten-fold CV; ACC mean±sd, forgetting mean). Reverse = ESCA→RCC→BRCA→LUNG; forward = the opposite. Fold-paired win counts are given in the text. Last two rows: the strongest published method, as reported in [9] and as rerun by us with its released code.

Configuration	Reverse		Forward	
	ACC↑	Forg.↓	ACC↑	Forg.↓
Selector fine-tuning (no preservation)	.452±.071	.601	.529±.077	.503
+ replay of snapshots	.821±.036	.114	.841±.019	.064
+ distillation + utility hinge (flat)	.830±.029	.090	.841±.043	.055
+ hierarchical selector (PathSelect)	.854±.044	.060	.811±.029	.061
Frozen head, mean pooling (no learned selection)	.679±.032	—	—	—
QPMIL-VL (reported)	.859±.032	.064	.890±.021	.027
QPMIL-VL (reproduced, released code)	.868±.046	.064	.884±.033	.038

orders are reported.

Table 3. **Preservation-component ablation with selection-level metrics** (flat, fold 1, 5-seed means). Replay restores old-task competence, the utility-preservation hinge preserves functional utility, and distillation stabilizes selection behavior; no single term or pair matches the full set. Jaccard and ΔU are reported for the lifecycle arms.

Arm	Rep.	Dist.	Util.	Class-IL↑	Task-IL↑	Leak.↓	Jaccard↑	ΔU ↑
Sequential fine-tuning (no LoRA)				.477	.809	.441	.002	−220
LoRA merge only				.447	.811	.476	.001	−301
		✓		.600	.853	.323	—	—
			✓	.809	.885	.091	—	—
	✓			.778	.907	.104	.086	−21.2
	✓	✓		.797	.897	.117	.175	−31.9
Full preservation (flat)	✓	✓	✓	**.824**	**.915**	**.101**	.161	**−16.8**

Do the three preservation losses do different jobs? (Table 3) Distillation alone leaves leakage at 0.323, three times the full method—in one seed 59 of 76 renal slides land in the lung classes—so distillation preserves *how* the selector scores but not *which task* the evidence belongs to. Replay (0.104) and the utility hinge (0.091) each restore task attribution on their own; among single terms the hinge reaches the highest class-IL (0.809) and replay the highest task-IL (0.907). In the ten-fold chain of Table 2, where components are added in order, replay accounts for most of the accuracy recovery, and the hinge's increment on top of replay and distillation is order-sensitive, as reported above. Under the hierarchy, removing the group level of distillation costs 3.95 task-IL and 3.71 class-IL points (5/5), so quota-level and patch-level preservation act at different resolutions. No single loss or pair matches the full set.

What does accuracy alone not show? (Table 3, right) Unprotected selectors retain almost none of their original evidence (Jaccard $0.0023/0.0010$) and lose old-task utility by an order of magnitude more than the fully preserved selector (ΔU $-220/-301$ vs. -16.8). LoRA merging alone shows no systematic difference from full fine-tuning (3/5): merging is the deployment substrate, not the preservation mechanism. Distillation without the hinge retains slightly more identity (0.1752 vs. 0.1613) yet is less accurate and forgets more: preserving *where* the policy looks does not secure *how useful* what it finds is—the gap the utility hinge closes. Three further interventions raised Jaccard without matching the full method's diagnostic performance: warm-starting task 1 with full-parameter training, damping each fold-in to half, and composing independently trained residuals post hoc.

Are the update-protocol choices necessary? Each probe was decided by a rule fixed before the run. Warm start and damped merging showed no systematic benefit, and a single adapter never reset lost 3.54 class-IL points on the flat substrate (4/5), so fresh per-task adapters are retained. Post-hoc composition refuted a pre-specified expectation: independently trained residuals, summed once, beat the bare sequential substrate on every metric in every seed (cf. task arithmetic [28]), yet trailed the full method by 13.1 points (5/5) because independence excludes replay and distillation; porting MergeSlide's orthogonal-projection merging [12] to our selector did not close this gap ($0.709\pm.035$). With 128 memory entries PathSelect still outperformed replay with 512 (task-IL $+3.06$, 5/5). A task-conditioned query, a stateful multi-round selector, and a group-level semantic prior failed their pre-specified criteria and were removed.

5. DISCUSSION AND LIMITATIONS

PathSelect suggests that continual learning in budgeted perception should distinguish two failure modes that accuracy alone conflates: a policy may change *where* it looks, and the new evidence may or may not remain *functionally equivalent*. The quantities in this paper therefore play different roles and should not be collapsed into one notion. Distillation preserves a soft behavior distribution, whereas Jaccard measures hard identity after the fact; candidate utility u_i teaches which patch is useful now, whereas set utility U evaluates whether the selected evidence remains useful across time; and the utility-preservation hinge uses U as a lower bound, deliberately allowing better alternative evidence. The study also shows what is *not* sufficient: LoRA merging alone does not protect the selector, and warm start, damped merging, and post-hoc composition each increase selection Jaccard without matching the full method's diagnostic performance (Sec. 4.3), so "choose the same patches" is not a complete objective—identity and function must be measured separately, which is why behavioral and functional preservation are treated as complementary constraints rather than redundant regularizers.

Our claims are bounded in three ways. First, the benchmark uses four organ-separated, highly separable TCGA tasks (a linear probe separates them at 98.21%), one frozen foundation model, one main budget ($B = 8$), known task boundaries during training, and five seeds or ten folds, so harder within-organ or mixed-task streams may yield different conclusions. Second, the hierarchical substrate's advantage and the accuracy increment of distillation and the utility hinge over replay are order-dependent (Table 2), while their forgetting reduction holds in 8/10 folds in both orders; pre-specified comparisons were made on the canonical reverse order and forward-order results are reported as measured. Third, the frozen head assumes the full class vocabulary as text from the start, the memory stores no image or feature tensors but replay requires access to the feature store and no formal privacy guarantee is claimed, and the fold-to-fold spread of our runs and of the reproduced baseline (sd 0.03–0.05) exceeds the gaps between the top methods. PathSelect is a one-shot evidence-acquisition policy; sequential, stateful acquisition on its task-free selector is left to future work.

6. CONCLUSION

PathSelect reframes continual WSI learning around an under-measured question: after learning new diagnostic tasks, does the model still know where to look? Freezing the diagnostic pathway isolates the evidence selector as the continual-learning object, and one shared selector is preserved at two levels—distillation maintains selection behavior, while a utility-preservation hinge lets evidence identity change without letting usefulness regress. On a public four-cohort benchmark the method is competitive with the strongest published result while adapting 16.4k selector parameters per task, and selection-level measurements show that evidence identity and utility drift differently. Continual learning for budgeted perception should therefore preserve not only what a model predicts, but also the policy by which it acquires the evidence used to predict.

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Compliance with Ethical Standards

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